

Chapter 2: Z-Transform

Introduction:

Transform techniques are an important tool in the analysis of signals and linear time invariant (LTI) systems.

The Z-transform plays the same role in the analysis of discrete-time signals & LTI systems as the Laplace transform does in the analysis of continuous-time signals & LTI systems. Laplace transform can be developed as an extension of the continuous-time Fourier transform and Z-transform as an extension of the discrete-time Fourier transform. This extension was motivated in part by the fact that it can be applied to a broader class of signals than the Fourier transform does not converge but the Laplace transform or Z-transform does. The Laplace transform and Z-transform allowed us, for example to perform transform analysis of unstable systems & to develop additional insights and tools for LTI system analysis.

The Direct Z-Transform:

The Z-transform of a discrete-time signal $x[n]$ is defined as an infinite sum or infinite power series

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

where z is a continuous complex variable. This expression is generally referred to as the two-sided z-transform.

For convenience, the z-transform of a signal $x[n]$ is denoted by

$$X(z) \equiv Z\{x[n]\}$$

Whereas the relationship between $x[n]$ and $X(z)$ is indicated by

$$x[n] \xleftrightarrow{z} X(z)$$

Since the z-transform is an infinite power series, it exists only for those values of z for which this series converges. The region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value. Thus, any time we cite a z-transform we should also indicate its ROC.

Find z-transform of following signals:

1. $x_1[n] = \{1, 2, 1, 3, 4\}$ [Causal Signal]
2. $x_2[n] = \{1, 2, 1, 3, 4, 0\}$ [Anti-causal signal]
3. $x_3[n] = \{1, 2, 1, 3, 4\}$ [Non-causal signal]

Z-Transform of some elementary signals:

1. $x[n] = \delta[n]$.

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n]z^{-n} = z^0 = 1 \quad \text{ROC: All } z.$$

2. $x[n] = \delta[n-k]$.

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n-k]z^{-n} = z^{-k} \quad \text{ROC: All } z \text{ except at } z = 0.$$

3. $x[n] = u[n]$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} u[n]z^{-n} \\ &= \sum_{n=0}^{\infty} z^{-n} \\ &= \frac{1}{1-z^{-1}} \text{ if } |z^{-1}| < 1 \\ &= \frac{1}{1-z^{-1}}, \quad \text{ROC: } |z| > 1 \end{aligned}$$

4. $x[n] = u[-n-1]$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} u[-n-1]z^{-n} \\ &= \sum_{n=-\infty}^{-1} z^{-n} \\ &= \sum_{n=1}^{\infty} z^n \\ &= \frac{z}{1-z} \text{ if } |z| < 1 \\ &= -\frac{1}{1-z^{-1}}, \quad \text{ROC: } |z| < 1 \end{aligned}$$

Analysis of ROC:

We can express the complex variable z in polar form as $z = re^{j\theta}$.

Then,

$$\begin{aligned} X(z)|_{z=re^{j\theta}} &= \sum_{n=-\infty}^{\infty} x[n](re^{j\theta})^{-n} \\ &= \sum_{n=-\infty}^{\infty} \{x[n]r^{-n}\}e^{-j\theta n} \end{aligned}$$

The region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value. In the ROC of $X(z)$,

$$\begin{aligned} |X(z)| &< \infty, \text{ But} \\ |X(z)| &= \left| \sum_{n=-\infty}^{\infty} \{x[n]r^{-n}\}e^{-j\theta n} \right| \\ &\leq \sum_{n=-\infty}^{\infty} |\{x[n]r^{-n}\}e^{-j\theta n}| \leq \sum_{n=-\infty}^{\infty} |x[n]r^{-n}| \end{aligned}$$

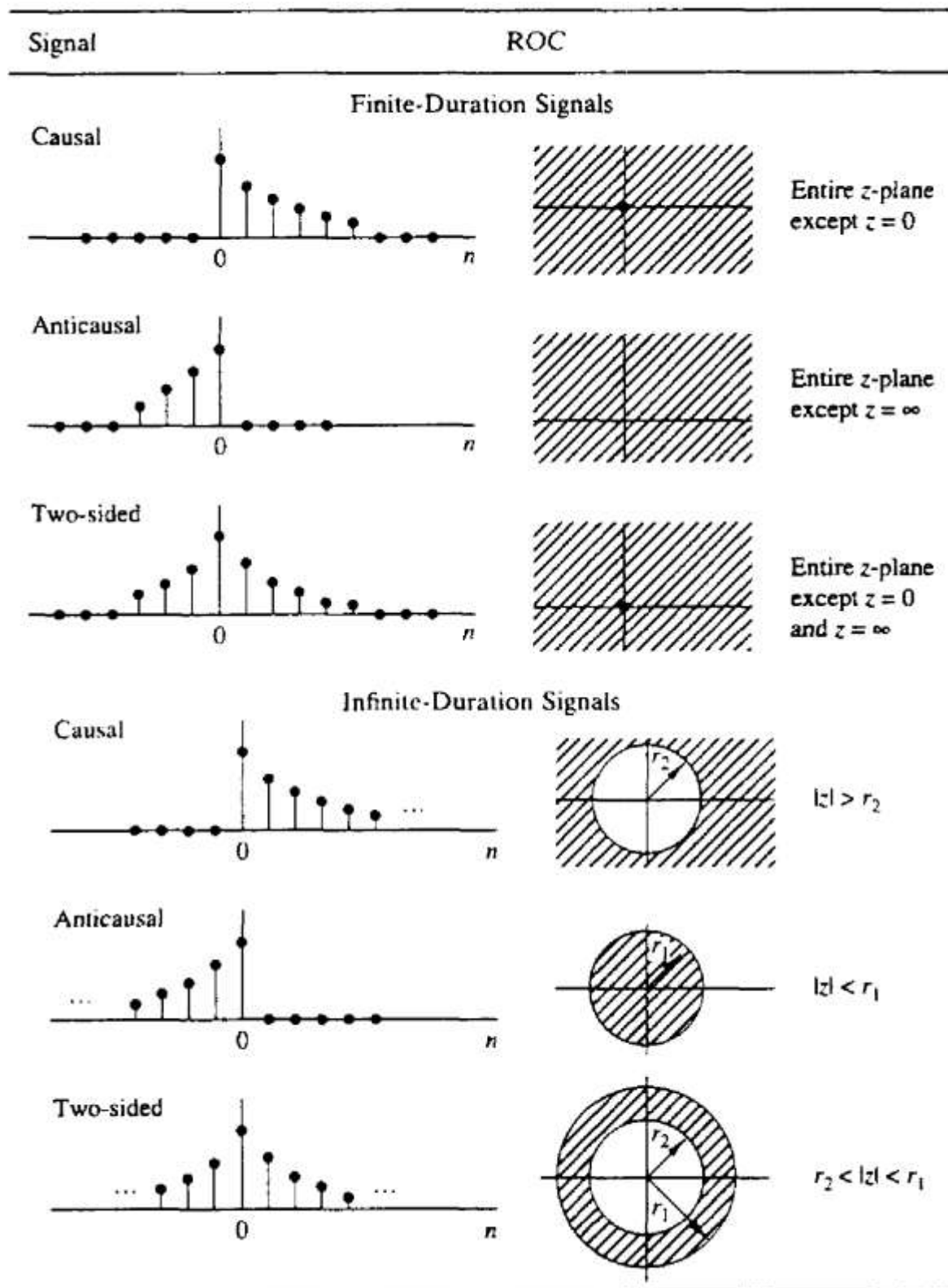


Figure : Characteristic families of signals with their corresponding ROC.

Hence $|X(z)|$ is finite if the sequence $x[n]r^{-n}$ is absolutely summable.
 To elaborate again,

$$|X(z)| \leq \sum_{n=-\infty}^{-1} |x[n]r^{-n}| + \sum_{n=0}^{\infty} |x[n]r^{-n}|$$

$$\leq \sum_{n=1}^{\infty} |x[-n]r^n| + \sum_{n=0}^{\infty} |x[n]r^{-n}|$$

The first series, a non-causal sequence, converges for $|z| < r_2$, and the second series, a causal sequence, converges for $|z| > r_1$, resulting in an annular region of convergence.

Properties of Z-transform:

1. Linearity:

If $x_1[n] \xrightarrow{z} X_1(z)$ & $x_2[n] \xrightarrow{z} X_2(z)$, then

$$x[n] = ax_1[n] + bx_2[n] \xrightarrow{z} aX_1(z) + bX_2(z)$$

For any constants a and b. ROC is intersection of the ROC of $X_1(z)$ and $X_2(z)$.

Find z-transform of $x[n] = (\cos\omega_0 n)u[n]$. (hint: use Euler's identity)

Ans:

$$X(z) = \frac{(1 - z^{-1}\cos\omega_0)}{1 - 2z^{-1}\cos\omega_0 + z^{-2}} \quad \text{ROC: } |z| > 1$$

2. Time Shifting:

If $x[n] \xrightarrow{z} X(z)$ then $x[n - k] \xrightarrow{z} z^{-k}X(z)$

ROC of $z^{-k}X(z)$ is the same as that of $X(z)$ except for $z = 0$ if $k > 0$ and $z = \infty$ if $k < 0$.

The properties of linearity and time shifting are the key features that make the z-transform extremely useful for the analysis of discrete-time LTI systems.

3. Scaling in z-domain:

If $x[n] \xrightarrow{z} X(z)$ ROC: $r_1 < |z| < r_2$

Then $a^n x[n] \xrightarrow{z} X(a^{-1}z)$ ROC: $|a|r_1 < |z| < |a|r_2$

for any constant a, real or complex.

Proof:

$$Z\{a^n x[n]\} = \sum_{n=-\infty}^{\infty} a^n x[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n](a^{-1}z)^{-n} = X(a^{-1}z)$$

ROC is $r_1 < |a^{-1}z| < r_2 \Rightarrow |a|r_1 < |z| < |a|r_2$

Eg:- $x[n] = a^n(\cos\omega_0 n)u[n]$

$$\text{We know, } Z\{\cos\omega_0 n u[n]\} = \frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}}, \quad \text{ROC: } |z| > 1$$

$$\text{Then, } Z\{a^n(\cos\omega_0 n) u[n]\} = \frac{1 - az^{-1}\cos\omega_0}{1 - 2az^{-1}\cos\omega_0 + (az^{-1})^2}, \quad \text{ROC: } |z| > |a|$$

Find z - transform of $x[n] = a^n(\cos\omega_0 n)u[n]$

4. Time Reversal:

$$\begin{aligned} \text{If } x[n] &\xrightarrow{z} X(z) & \text{ROC: } r_1 < |z| < r_2 \\ \text{Then, } x[-n] &\xrightarrow{z} X(z^{-1}) & \text{ROC: } \frac{1}{r_2} < |z| < \frac{1}{r_1} \end{aligned}$$

Proof:

$$Z\{x[-n]\} = \sum_{n=-\infty}^{\infty} x[-n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n](z^{-1})^{-n} = X(z^{-1})$$

$$\text{ROC: } r_1 < |z^{-1}| < r_2 \implies \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

Eg:- $x[n] = a^n u[-n]$

Ans: $\frac{1}{1-az}$, ROC: $|z| < |a|$

5. Differentiation in the z-domain:

$$\begin{aligned} \text{If } x[n] &\xrightarrow{z} X(z) & \text{ROC: } r_1 < |z| < r_2 \\ \text{Then, } nx[n] &\xrightarrow{z} -z \frac{dX(z)}{dz} & \text{ROC: } r_1 < |z| < r_2 \end{aligned}$$

Proof:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Differentiating both sides w.r.t. z

$$\frac{d}{dz} X(z) = \sum_{n=-\infty}^{\infty} x[n](-n)z^{-n-1} = -z^{-1} \sum_{n=-\infty}^{\infty} \{nx[n]\}z^{-n} = -z^{-1} Z\{nx[n]\}$$

Eg. Determine the signal $x[n]$ whose z-transform is given by

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|, \quad \text{Ans: } nx[n] = (-1)^{n+1} a^n u[n-1]$$

$x[n] = n.a^n u[n]$

6. Convolution of two Sequences:

$$\text{If, } x_1[n] \xrightarrow{z} X_1(z) \text{ \& } x_2[n] \xrightarrow{z} X_2(z)$$

$$\text{then, } x[n] = x_1[n] * x_2[n] \xrightarrow{z} X(z) = X_1(z)X_2(z) \quad \text{ROC: } R_{x_1} \cap R_{x_2}$$

Proof:

$$x[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]$$

$$X(z) = Z\{x_1[n] * x_2[n]\} = \sum_{n=-\infty}^{\infty} \{x_1[n] * x_2[n]\}z^{-n} = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]z^{-n}$$

Upon interchanging the order of summation

$$X(z) = \sum_{k=-\infty}^{\infty} x_1[k] \sum_{n=-\infty}^{\infty} x_2[n-k]z^{-n}$$

Now, applying time shifting property

$$X(z) = \sum_{k=-\infty}^{\infty} x_1[k]z^{-k}X_2(z) = X_1(z)X_2(z)$$

Computation of the convolution of two signals $x_1[n]$ & $x_2[n]$ using the z-transform:

- (1) Find $X_1(z)$ & $X_2(z)$.
- (2) Multiply $X_1(z)$ & $X_2(z)$.
- (3) Find the inverse z-transform of multiplication.

This procedure is, in many cases, computationally easier than the direct evaluation of the convolution summation.

7. Correlation of two Sequences:

$$\text{If, } x_1[n] \xleftrightarrow{z} X_1(z) \text{ \& } x_1[n] \xleftrightarrow{z} X_1(z)$$

$$\text{then, } r_{x_1x_2}(l) = \sum_{n=-\infty}^{\infty} x_1[n]x_2[n-l] \xleftrightarrow{z} R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$$

$$\text{ROC: } -\text{ROC of } X_1(z) \cap \text{ROC of } X_2(z^{-1})$$

Proof:

$$r_{x_1x_2}(l) = x_1[l] * x_2[-l]$$

Using convolution property

$$R_{x_1x_2}(z) = Z\{x_1(l)\} \cdot Z\{x_2(l)\}$$

Using time-reversal property

$$R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$$

8. Conjugate of a Complex Sequence:

$$x^*[n] \xleftrightarrow{z} X^*(z^*) \quad \text{ROC: } \text{ROC } X(z)$$

9. Multiplication of two sequences:

$$\text{If, } x_1[n] \xleftrightarrow{z} X_1(z) \text{ \& } x_1[n] \xleftrightarrow{z} X_1(z)$$

$$\text{then, } x[n] = x_1[n]x_2[n] \xleftrightarrow{z} X(z) = \frac{1}{2\pi j} \oint_c X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$$

Where c is a closed contour that encloses the origin & lies within the ROC common to both $X_1(v)$ & $X_2(1/v)$.
 ROC: $r_{1l}r_{2l} < |z| < r_{1u}r_{2u}$

10. Parseval's Relation:

If $x_1[n]$ and $x_2[n]$ are complex-valued sequences, then

$$\sum_{n=-\infty}^{\infty} x_1[n]x_2^*[n] = \frac{1}{2\pi j} \oint_c X_1(v)X_2^*\left(\frac{1}{v^*}\right)v^{-1}dv$$

Provided that $r_{1l}r_{2l} < 1 < r_{1u}r_{2u}$, where $r_{1l} < |z| < r_{1u}$ and $r_{2l} < |z| < r_{2u}$ are the ROC of $X_1(z)$ and $X_2(z)$.

11. Initial Value Theorem:

If $x[n]$ is causal [i.e. $x[n] = 0$ for $n < 0$], then

$$x[0] = \lim_{z \rightarrow \infty} X(z)$$

Proof:

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

$$\lim_{z \rightarrow \infty} X(z) = \lim_{z \rightarrow \infty} x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots = x[0]$$

12. Final Value Theorem:

If $x[n]$ is causal, then

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow \infty} (z-1)X(z)$$

Note: All the poles of $X(z)$ should lie inside the unit circle.

Inversion of Z-Transform:

1. Direct evaluation by Contour integration (not in syllabus).
2. Expansion into a series of terms in the variables z and z^{-1} .
3. Partial fraction expansion and table lookup.

Long-Division Method:

Determine the inverse z-transform of

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

When (a) ROC: $|z| > 1$. (b) ROC: $|z| < 0.5$.

Solution: (a) Since the ROC is the exterior of a circle, we expect $x[n]$ to be a causal signal.

Thus we seek a power series expansion in negative powers of z .

$$X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}$	$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}$	$1 - \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \dots$
	$\frac{3}{2}z^{-1} - \frac{1}{2}z^{-2}$ $\frac{3}{2}z^{-1} - \frac{9}{4}z^{-2}$ $\quad + \frac{3}{4}z^{-3}$	

$$X(z) = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \dots$$

$$\therefore x[n] = \left\{ \underset{\uparrow}{1}, \frac{3}{2}, \frac{7}{4}, \frac{15}{8}, \dots \right\}.$$

(b) In this case ROC is the interior of a circle. The signal $x[n]$ is anticausal. So, we seek a power expansion in powers of z , so we perform the long division in the following way:

$$X(z) = 2z^2 + 6z^3 + 14z^4 + \dots$$

$$\therefore x[n] = \left\{ \dots, 30, 14, 6, 2, 0, \underset{\uparrow}{0} \right\}$$

This method is used only if one wished to determine the values of the first few samples of the signal.

The Inverse Z-Transform by Partial Fraction Expansion:

In the table lookup method, we attempted to express the function $X(z)$ as a linear combination.

$$X(z) = \alpha_1 X_1(z) + \alpha_2 X_2(z) + \dots + \alpha_k X_k$$

Then the inverse z-transform of $X(z)$, can be found using the linearity property as,

$$x[n] = \alpha_1 x_1[n] + \alpha_2 x_2[n] + \dots + \alpha_k x_k[n]$$

This approach is useful if $X(z)$ is a rational function,

$$\begin{aligned} X(z) &= \frac{N(z)}{D(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \dots (A) \\ &= \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}, \quad a_0 \cong 1 \end{aligned}$$

Note: If $a_0 \neq 1$, divide $N(z)$ & $D(z)$ by a_0 .

A rational function of the form (A) is called proper, if $a_N \neq 0$ and $M < N$ (finite zeros is less than the number of finite poles).

An improper rational function ($M \geq N$) can always be written as the sum of a polynomial & a proper rational function.

$$\text{Let } X(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

Where $a_N \neq 0$ and $M < N$

Eliminating negative powers of z

$$X(z) = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \dots + a_N}$$

Since $N > M$, the function

$$\frac{X(z)}{z} = \frac{b_0 z^{N-1} + b_1 z^{N-2} + \dots + b_M z^{N-M-1}}{z^N + a_1 z^{N-1} + \dots + a_N}$$

Is also always proper.

Now the next step is to express $X(z)/z$ as a sum of simple fractions.

Two cases:

(I) Distinct Poles:

Suppose that the poles $p_1, p_2, \dots, p_N$ are all different (distinct)

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \dots + \frac{A_N}{z - p_N}$$

The problem is to determine the coefficients $A_1, A_2, \dots, A_N$.

$$A_k = \left. \frac{(z - p_k)X(z)}{z} \right|_{z = p_k}, \quad k = 1, 2, 3, \dots, N$$

(II) Multiple order poles:

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \frac{A_3}{(z - p_2)^2}$$

$$A_1 = \left. \frac{(z - p_1)X(z)}{z} \right|_{z = p_1}$$

$$A_3 = \left. \frac{(z - p_2)^2 X(z)}{z} \right|_{z = p_2}$$

$$A_2 = \left. \frac{d}{dz} \frac{(z - p_2)^2 X(z)}{z} \right|_{z = p_2}$$

In terms of z^{-1}

$$X(z) = \frac{A_1}{1 - p_1 z^{-1}} + \frac{A_2}{1 - p_2 z^{-1}} + \frac{A_3}{(1 - p_2 z^{-1})^2}$$

$$A_1 = (1 - p_1 z^{-1})X(z) \Big|_{z = p_1}$$

$$A_3 = (1 - p_2 z^{-1})^2 X(z) \Big|_{z = p_2}$$

$$A_2 = z \cdot \frac{d}{dz} (1 - p_2 z^{-1})^2 X(z) \Big|_{z = p_2}$$

Determine the z-transform of

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

If

(a) ROC: $|z| > 1$

(b) ROC: $|z| < 0.5$

(c) ROC: $0.5 < |z| < 1$.

Solution: $\rightarrow$

$$\begin{aligned} X(z) &= \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}} = \frac{1}{1 - z^{-1} - 0.5z^{-1} + 0.5z^{-2}} = \frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})} \\ &= \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.5z^{-1}} \end{aligned}$$

Solving, we get $A = 2$ and $B = -1$.

$$\text{Ans (a): } x[n] = 2u[n] - 0.5^n u[n]$$

$$\text{Ans (b): } x[n] = -2u[-n - 1] + 0.5^n u[-n - 1]$$

$$\text{Ans (c): } x[n] = -2u[-n-1] - 0.5^n u[n]$$

Causality and Stability:

Causality: -

We know, a causal LTI system is one whose unit sample response satisfies the condition

$$h[n] = 0 \quad n < 0$$

We also know that the ROC of the z-transform of a causal sequence is the exterior of a circle. Consequently, a linear time invariant system is a causal if and only if the ROC of the system function is the exterior of a circle of radius $r < \infty$, including the point $z = \infty$.

Stability: -

A necessary and sufficient condition for a LTI system to be BIBO stable is,

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

In turn, this condition implies that $H(z)$ must contain the unit circle within its ROC.

Indeed since

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

It follows that,

$$\begin{aligned} |H(z)| &\leq \sum_{n=-\infty}^{\infty} |h[n]z^{-n}| \\ &\leq \sum_{n=-\infty}^{\infty} |h[n]| |z^{-n}| \end{aligned}$$

When evaluated on the unit circle i.e. $|z| = 1$

$$|H(z)| \leq \sum_{n=-\infty}^{\infty} |h[n]|$$

Hence, if the system is BIBO stable, the unit circle is contained in the ROC of $H(z)$. The converse is also true. Therefore, a linear time invariant system is BIBO stable if and only if the ROC of the system function includes the unit circle.

A causal linear time invariant system (ROC: $|z| > r < 1$) is BIBO stable if and only if all the poles of $H(z)$ are inside the unit circle.

Example

A LTI system is characterized by the system function,

$$H(z) = \frac{3 - 4z^{-1}}{1 - 3.5z^{-1} + 1.5z^{-2}}$$

Specify the ROC of $H(z)$ and determine $h[n]$ for the following conditions:

- (a) The system is stable.
- (b) The system is causal.
- (c) The system is anticausal.

Solution →

The given system is

$$H(z) = \frac{3 - 4z^{-1}}{1 - 3.5z^{-1} + 1.5z^{-2}} = \frac{1}{1 - 1/2 z^{-1}} + \frac{2}{1 - 3z^{-1}}$$

The system has poles at $z = 1/2$ and $z = 3$.

(a) Since the system is stable, its ROC must include the unit circle & hence it is $1/2 < |z| < 3$.

Consequently $h[n]$ is non-causal and is given as,

$$h[n] = (1/2)^n u[n] - 2(3)^n u[-n-1].$$

(b) Since the system is causal, its ROC is $|z| > 3$. In this case,

$$h[n] = (1/2)^n u[n] + 2(3)^n u[n].$$

This system is unstable.

(c) If the system is anticausal, its ROC is $|z| < 0.5$ hence

$$h[n] = -[(1/2)^n + 2(3)^n] u[-n-1].$$

In this case the system is unstable.

Partial Fraction Expansion for multiple order poles:

$$\frac{X(z)}{z} = \sum_p \frac{A_p}{z - z_p} + \sum_q \sum_{k=1}^{M_q} \frac{B_{qk}}{(z - z_q)^k}$$

Where z_p is the p^{th} single pole and z_q is the q^{th} multiple pole of the order M_q . The constants A_p and B_{qk} are given as follows:

$$A_p = (z - z_p) X(z) \Big|_{z = z_p}$$

$$B_{qk} = \frac{1}{(M_q - k)!} \cdot \frac{d^{M_q - k}}{dz^{M_q - k}} (z - z_q)^{M_q} X(z) \Big|_{z = z_q}$$

	Signal, $x(n)$	z -Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3	$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
4	$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
5	$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
6	$-na^n u(-n - 1)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
7	$(\cos \omega_0 n) u(n)$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
8	$(\sin \omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
9	$(a^n \cos \omega_0 n) u(n)$	$\frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
10	$(a^n \sin \omega_0 n) u(n)$	$\frac{az^{-1} \sin \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Figure : Common Z-transform pairs.

Reference:

1. J. G. Proakis, D. G. Manolakis, "Digital Signal Processing, Principles, Algorithms and Applications", 3rd Edition, Prentice-hall, 2000. Chapter 3.